

Numerical and Algebraic Expressions — Distributing and Combining Without Breaking the Rules

Two rules do most of the work on this page, and breaking either one is the most common way to lose an expression problem.

- **Distributing** means multiplying *every* term inside the parentheses — including the sign in front of each one.
- **Like terms** have exactly the same letter part. $5x$ and $2x$ are like; $5x$ and $3y$ are not, and neither are $5x$ and 5 .
- **The substitution test:** if two expressions are really equivalent, they give the same value for every number you put in. One disagreement proves they are different expressions.

1. Write $4(x + 7)$ without parentheses. Multiply every term inside by 4.

Answer: _____

2. **Find and fix.** Kim writes $3(x + 5) = 3x + 5$.

Test both expressions at $x = 4$. Kim's version gives 17. Work out the value of $3(x + 5)$ at $x = 4$ and write it on the answer line, then say in a few words which term Kim forgot to multiply.

Answer: _____

3. Find and fix. Devon writes $5x + 3y = 8xy$.

Test both expressions at $x = 2$ and $y = 3$. Devon's version gives 48. Work out the value of $5x + 3y$ at those values and write it on the answer line, then say in a few words why $5x$ and $3y$ cannot be pulled into one term.

Answer: _____

4. Simplify by combining like terms only:

$$7a + 4 - 2a + 9$$

Answer: _____

5. Find and fix. Simplify $8 - 2(x - 3)$.

Nadia writes $8 - 2x - 6$, which is $2 - 2x$, and at $x = 1$ her version gives 0.

The -2 must be multiplied by *both* terms in the parentheses. Work out the correct value of $8 - 2(x - 3)$ at $x = 1$ and write it on the answer line.

Answer: _____

6. Simplify completely:

$$5(2x - 3) + 4x$$

Distribute first, then combine only the terms that are alike.

Answer: _____

7. Factor $6x + 15$ by taking out the greatest common factor. Then check your answer by distributing it back — the check should return $6x + 15$ exactly.

Answer: _____

8. Explain, then simplify.

- (a) A classmate insists that $5x + 3y$ “must be able to become one term.” Explain, in your own words, why two unlike terms cannot be combined. Remember that a letter stands for a number nobody has told you yet.
- (b) Simplify $3(2x - 4) - 2(x - 5)$, then check your simplified expression by substituting $x = 3$ into it. Write that value on the answer line. (A classmate who keeps both signs inside the second bracket gets -10 here.)

Answer: _____